Android Studio Lab Experiment

Aim: Write a program implementing basic arithmetic operations of a simple calculator.

1. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/number1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number 01"
        android:inputType="numberDecimal" />

    <EditText
        android:id="@+id/number2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:hint="Enter Number 02"
        android:inputType="numberDecimal" />

    <TextView
        android:id="@+id/result_text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textColor="@color/black"
        android:textSize="17sp"
        android:textStyle="bold" />

    <Button
        android:id="@+id/add_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
        android:layout_marginTop="10dp"
        android:text="+" />

    <Button
        android:id="@+id/sub_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="-" />

    <Button
        android:id="@+id/mul_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="*" />

    <Button
        android:id="@+id/div_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="/" />

    <Button
        android:id="@+id/clear_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="clear" />

</LinearLayout>
```

2. MainActivity.java

```
package com.example.calculator;


import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
```

```
    EditText number1, number2;
    TextView result_text;
    Button add_btn, sub_btn, mul_btn, div_btn, clear_btn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        number1 = findViewById(R.id.number1);
        number2 = findViewById(R.id.number2);
        result_text = findViewById(R.id.result_text);
        add_btn = findViewById(R.id.add_btn);
        sub_btn = findViewById(R.id.sub_btn);
        mul_btn = findViewById(R.id.mul_btn);
        div_btn = findViewById(R.id.div_btn);
        clear_btn = findViewById(R.id.clear_btn);

        add_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String number1_text = number1.getText().toString();
                String number2_text = number2.getText().toString();
                int num1 = Integer.parseInt(number1_text);
                int num2 = Integer.parseInt(number2_text);
                float sum = num1 + num2;
                result_text.setText("The addition of the two numbers is: " + sum);
                Toast.makeText(getApplicationContext(), "The addition of the two
numbers is: " + sum, Toast.LENGTH_SHORT).show();
            }
        });

        sub_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String number1_text = number1.getText().toString();
                String number2_text = number2.getText().toString();
                int num1 = Integer.parseInt(number1_text);
                int num2 = Integer.parseInt(number2_text);
                float sub = num1 - num2;
                result_text.setText("The subtraction of the two numbers is: " + sub);
                Toast.makeText(getApplicationContext(), "The subtraction of the two
numbers is: " + sub, Toast.LENGTH_SHORT).show();
            }
        });

        mul_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
```

```
                String number1_text = number1.getText().toString();
                String number2_text = number2.getText().toString();
                int num1 = Integer.parseInt(number1_text);
                int num2 = Integer.parseInt(number2_text);
                float mul = num1 * num2;
                result_text.setText("The multiplication of the two numbers is: " +
mul);
                Toast.makeText(getApplicationContext(), "The multiplication of the two
numbers is: " + mul, Toast.LENGTH_SHORT).show();
            }
        });

        div_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String number1_text = number1.getText().toString();
                String number2_text = number2.getText().toString();
                int num1 = Integer.parseInt(number1_text);
                int num2 = Integer.parseInt(number2_text);
                float div = (float) num1 / num2;
                result_text.setText("The division of the two numbers is: " + div);
                Toast.makeText(getApplicationContext(), "The division of the two
numbers is: " + div, Toast.LENGTH_SHORT).show();
            }
        });

        clear_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                number1.setText("");
                number2.setText("");
                result_text.setText("");
                Toast.makeText(getApplicationContext(), "Inputs cleared...",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```